DEPARTMENT OF AGRICULTURE IV-A · DEPDev IV-A

Food Insecurity Risk Forecast

Quezon · Q4 2025 (Oct-Dec)

Generated 29 September 2026 from aiPHeed model output.

What this number is

The modelled probability that this area falls in the HIGH class of the composite food stress label. **It is an indicative relative risk estimate, not an official food insecurity classification**, and it is published at the Maximum Reliable Lead Time the model established empirically.

Headline

FOOD INSECURITY RISK PROBABILITY

0.477

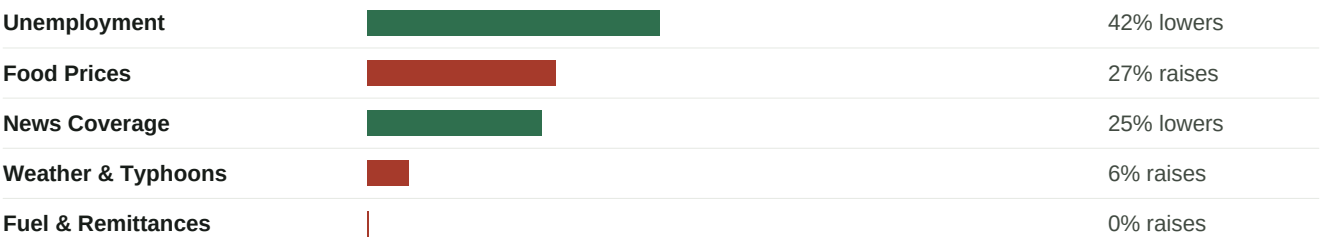
LOW · quarter-on-quarter +0.0000

Risk class	**LOW**
Rank in CALABARZON	**5 of 5**
Population (2020 CPH)	**2,229,383**
Poverty incidence (PSA 2021)	**16.6%**
Articles analysed	**0**

Forecast horizon

Same-quarter estimate. No forward lead time met the 75% reliability threshold of Section 3.4.4, so none is published.. Indicative relative risk only. Walk-forward accuracy did not reach 75% at any forward horizon, so this output is reported as a same-quarter estimate and not as a forecast.

What is driving the figure



Red bars raise the predicted shortfall, **green** lower it. A large bar is not automatically bad news -- share of the explanation and direction of effect are separate things.

For Quezon in 2025-Q4, the model's prediction breaks down as: Unemployment 42%, Food Prices 27%, News Coverage 25%, Weather & Typhoons 6%, Fuel & Remittances 0%. Unemployment is the largest single contributor, holding the score down. Food Prices clears the 20% line while raising the score. Unemployment and News Coverage clear the 20% line while lowering it. Shares are SHAP contributions to the province's food insecurity risk probability, recomputed every quarter.

Municipal distribution

Municipal values reweight the province figure by poverty incidence (60%) and population density (40%). **They are not independent municipal forecasts** -- every municipality in a province rises and falls with it.

#	City / Municipality	Class	Population	Poverty	Index
1	**Sariaya**	Municipality	161,868	18.7%	0.3228
2	**Tiaong**	Municipality	106,265	16.8%	0.2590
3	**San Antonio**	Municipality	35,891	16.8%	0.2569
4	**Plaridel**	Municipality	10,129	16.8%	0.2554
5	**Dolores**	Municipality	32,514	16.8%	0.2535
6	**Pagbilao**	Municipality	78,700	16.8%	0.2506
7	**Lucban**	Municipality	53,091	16.8%	0.2480
8	**Agdangan**	Municipality	12,764	16.8%	0.2479
9	**Gumaca**	Municipality	71,942	16.8%	0.2466
10	**Padre Burgos**	Municipality	23,488	16.8%	0.2447
11	**Pitogo**	Municipality	22,798	16.8%	0.2432
12	**Catanauan**	Municipality	72,752	16.8%	0.2421
13	**Alabat**	Municipality	15,936	16.8%	0.2415
14	**Atimonan**	Municipality	64,260	16.8%	0.2411
15	**Lopez**	Municipality	94,657	16.8%	0.2410
16	**General Luna**	Municipality	24,804	16.8%	0.2400
17	**Infanta**	Municipality	76,186	16.8%	0.2389
18	**Quezon**	Municipality	15,886	16.8%	0.2389
19	**Perez**	Municipality	12,767	16.8%	0.2389
20	**Macalelon**	Municipality	27,312	16.8%	0.2388
21	**Calauag**	Municipality	71,809	16.8%	0.2388

#	City / Municipality	Class	Population	Poverty	Index
22	**San Andres**	Municipality	37,454	16.8%	0.2386
23	**Candelaria**	Municipality	137,881	15.4%	0.2383
24	**Unisan**	Municipality	25,448	16.8%	0.2380
25	**Guinayangan**	Municipality	44,045	16.8%	0.2380
26	**San Francisco**	Municipality	62,097	16.8%	0.2380
27	**San Narciso**	Municipality	51,058	16.8%	0.2374
28	**Buenavista**	Municipality	31,160	16.8%	0.2374
29	**Mauban**	Municipality	71,081	16.8%	0.2363
30	**Jomalig**	Municipality	7,667	16.8%	0.2349
31	**Mulanay**	Municipality	55,576	16.8%	0.2344
32	**Sampaloc**	Municipality	13,629	16.8%	0.2343
33	**Polillo**	Municipality	31,908	16.8%	0.2341
34	**Burdeos**	Municipality	24,644	16.8%	0.2340
35	**Real**	Municipality	38,678	16.8%	0.2335
36	**Patnanungan**	Municipality	15,052	16.8%	0.2332
37	**Tagkawayan**	Municipality	54,003	16.8%	0.2329
38	**Panukulan**	Municipality	16,376	16.8%	0.2315
39	**General Nakar**	Municipality	34,225	16.8%	0.2292
40	**Lucena**	City	278,924	9.2%	0.1910
41	**Tayabas**	City	112,658	14.3%	0.1766

News evidence

No relevant articles were geocoded to this subject for the quarter.

Method

The model predicts, for each CALABARZON province, the probability that the province falls in the HIGH class of the composite food stress label -- SWS quarterly hunger prevalence combined with the province's food CPI deviation from the regional mean, thresholded at the global median. Inputs are the bias-corrected Food Stress Sentiment Index and its lags and acceleration, the five bilingual trigger proportions, and government indicators from PSA, BSP, DOE, PAGASA and PhilRice.

Measure	Value
Maximum Reliable Lead Time	0 quarter(s)
Accuracy	0.6000
Precision (weighted)	0.9200
Recall (weighted)	0.6000
F1 score (weighted)	0.6762
AUC-ROC	0.9444
F1 target (>= 0.75)	FAIL
AUC-ROC target (>= 0.80)	PASS
Features in the model	12
Baseline: aipheed lgbm combined	0.6727
Baseline: full feature logistic	0.5455
Baseline: lgbm nlp only	0.6182
Baseline: lgbm psa only	0.7273
Baseline: naive persistence	0.8000
Baseline: nlp only logistic	0.5455
Baseline: psa only logistic	0.5818

Accuracy must be read against its baselines. The comparison models above are evaluated on the same walk-forward folds at the same forecast gap, so the figures are directly comparable. Naive persistence in particular is a demanding baseline on an autocorrelated label: a model that does not clearly beat it is not earning its complexity.

Limitations

Indicative risk, not official classification

Outputs are relative ordinal estimates of potential risk. They are not definitive determinations of food insecurity status and must not be read as official government assessments or policy recommendations.

Proxy label, not a direct measurement

No publicly accessible province-quarter food insecurity label exists in the Philippines. The training target is a composite of SWS hunger prevalence, which has no province-level resolution, and PSA food CPI deviation, which reflects supply-side price conditions rather than household hunger experience directly.

Municipal figures are disaggregated, not modelled

Municipal values reweight the province figure by poverty incidence (60%) and population density (40%). Poverty values are largely province means rather than municipal Small Area Estimates, so within-province ordering is indicative only.

Reviewed before publication

Province-quarter figures pass through analyst review. A withheld figure is excluded from this report and from the regional average.

Scope of this assessment

This assessment reports findings only. It does not recommend interventions: no validated mapping exists between this indicator and a response protocol. Pair it with the Department's own standing guidance.